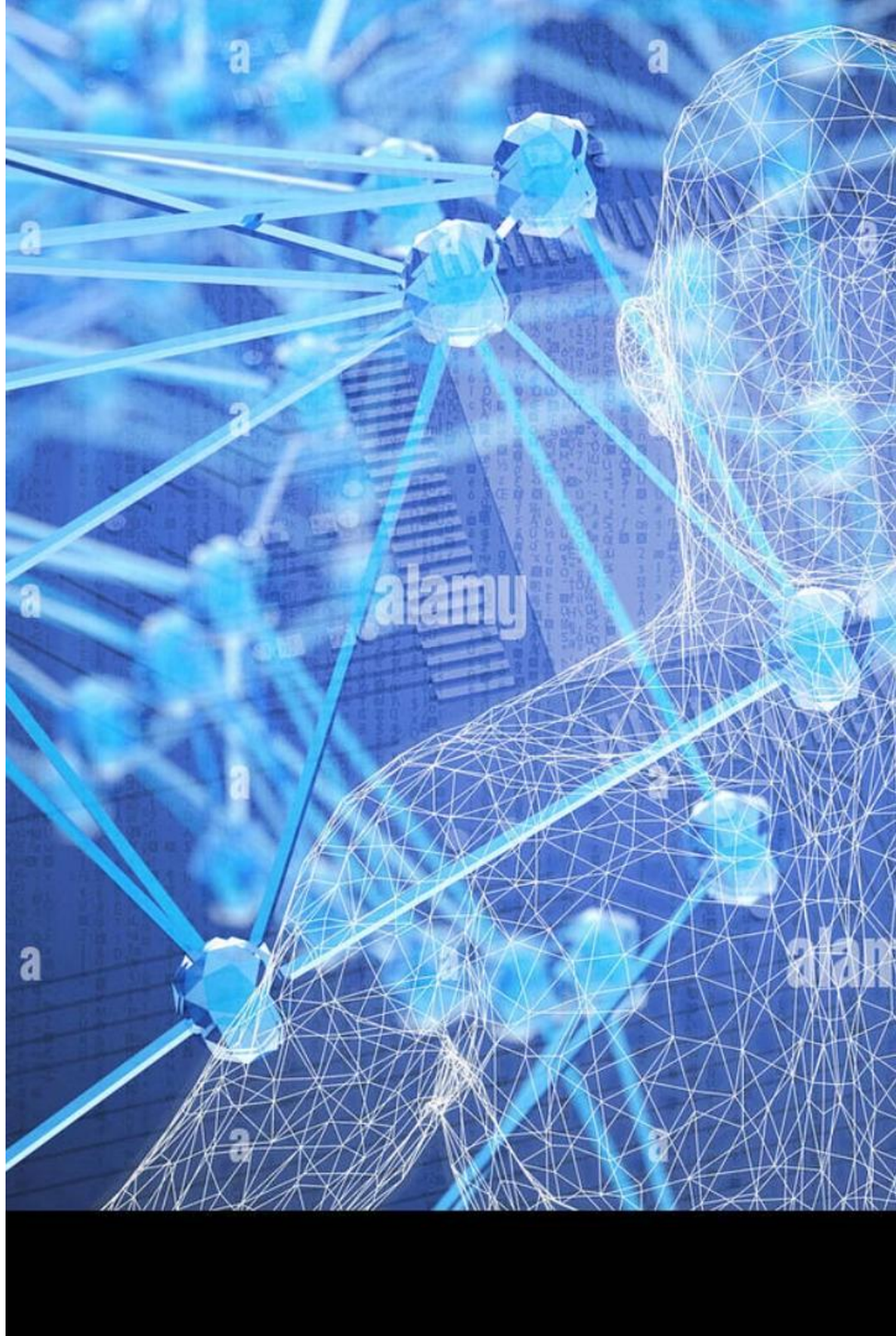
AI Capstone Project with Deep Learning

Subtitle: Coursera Certificate Presentation

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subject : Automata Computability and Complexity

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Course Overview

This course, offered by IBM through Coursera, is part of the IBM AI Engineering Professional Certificate. It provides hands-on experience with the full deep learning workflow. The primary goal is to help learners apply their deep learning knowledge in an end-to-end project, specifically an image classification task. The course supports both PyTorch and Keras/TensorFlow for implementation and uses Jupyter Notebooks for development.

Hands-on deep learning workflow

Full project lifecycle experience

Image classification task

Apply knowledge in a practical context

Supports PyTorch & Keras/TensorFlow

Flexible implementation frameworks

Development via Jupyter Notebooks

Interactive coding and visualization

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Module 1 - Loading Data

In this module, learners focused on the initial step of data acquisition and exploration. They were introduced to the process of downloading and organizing image data into training, validation, and test sets. Both PyTorch and Keras libraries were supported. The module emphasized basic visualization techniques and understanding dataset structure to lay the groundwork for model development.

Data Acquisition

Download and organize training, validation, test sets

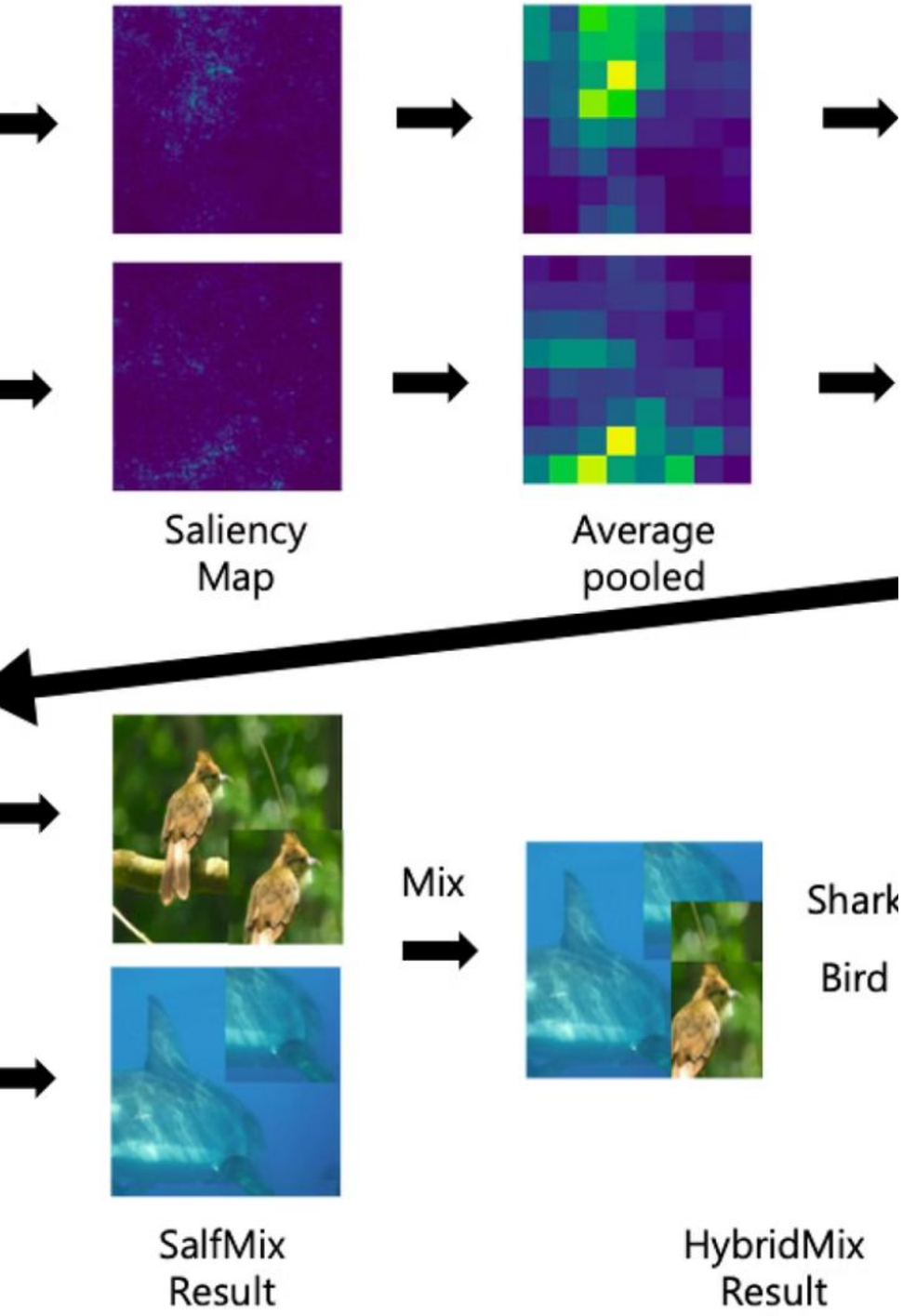
Library Support

PyTorch and Keras for loading data

Visualization

Basic exploratory techniques to understand data

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Module 2 - Data Preparation

This module guided learners in preparing image data for training deep learning models. It covered normalization and augmentation techniques, which help increase model accuracy and generalization. The tools included data loaders for PyTorch and data generators for Keras, enabling efficient handling of large datasets during training. This preparation was essential to ensure consistency and performance in model training.

Normalization

Standardize input for consistency

Augmentation

Enhance training data diversity

Data Loaders & Generators

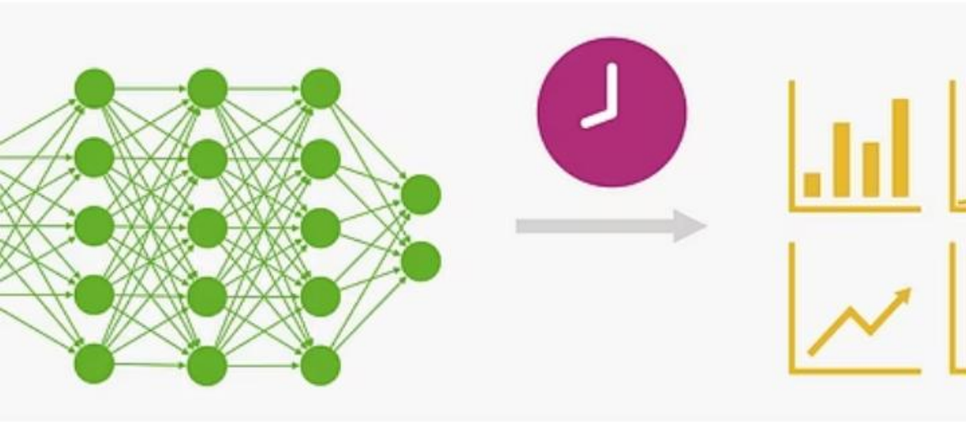
Enable efficient batch processing

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§6 **WHAT**

What data, features, and relationships in deep learning can be visualized?

Computational Graph & Network Architecture
Learned Model Parameters
Individual Computational Units
Neurons In High-dimensional Space
Aggregated Information



§7 **HOW**

How can we visualize deep learning data, features, and relationships?

Node-link Diagrams for Network Architecture
Dimensionality Reduction & Scatter Plots
Line Charts for Temporal Metrics
Instance-based Analysis & Exploration
Interactive Experimentation
Algorithms for Attribution & Feature Visualizatic

Module 3 - Building and Training Models

Learners transitioned to constructing and training models using transfer learning. In PyTorch, they implemented a simple linear classifier, while in Keras, they utilized the ResNet50 pre-trained model. The focus was on designing model architectures, adjusting hyperparameters, and monitoring training progress using loss and accuracy metrics. This module solidified learners' understanding of the core mechanics of training deep learning models.

Transfer Learning

Leverage pre-trained networks

PyTorch

Linear classifier implementation

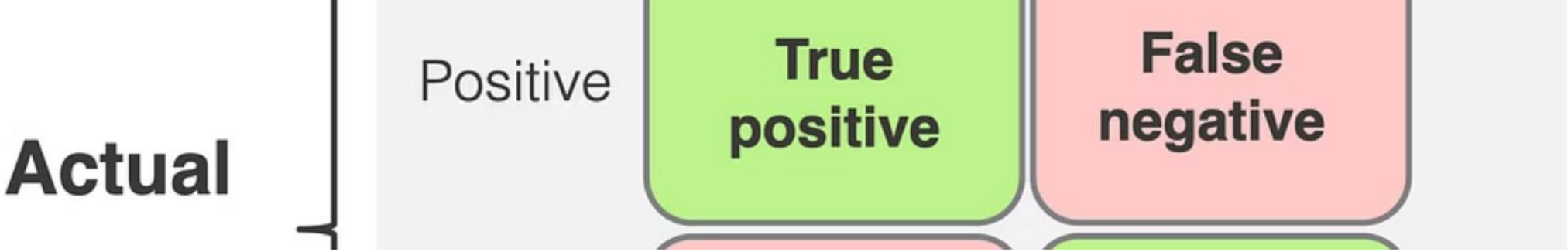
Keras

ResNet50 pre-trained model

Monitoring

Track loss and accuracy metrics

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Module 4 - Final Project and Evaluation

This capstone module required learners to build a complete image classification model using either PyTorch or Keras. PyTorch users implemented a classifier using ResNet18, while Keras users compared performance between ResNet50 and VGG16. The emphasis was on final evaluation, interpretation of results using metrics like accuracy and confusion matrices, and submitting a peer-reviewed final project that demonstrated the full deep learning pipeline.

1. PyTorch
   ResNet18 classifier
2. Keras
   Compare ResNet50 and VGG16
3. Evaluation Metrics
   Accuracy, confusion matrix analysis
4. Peer Review
   Submit and assess final project

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Project Results and Insights

The final model demonstrated robust performance, with accuracy metrics showing effective learning from the data. Learners visualized outcomes using confusion matrices and accuracy/loss plots. The experience offered valuable insights into challenges like overfitting, class imbalance, and optimization, which are common in real-world AI projects.

Performance

High accuracy on test data

Visualization

Confusion matrices and plots

Challenges

- Overfitting
- Class imbalance
- Optimization tuning

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Key Learnings

End-to-end Workflow

Loading, preprocessing, training, evaluation

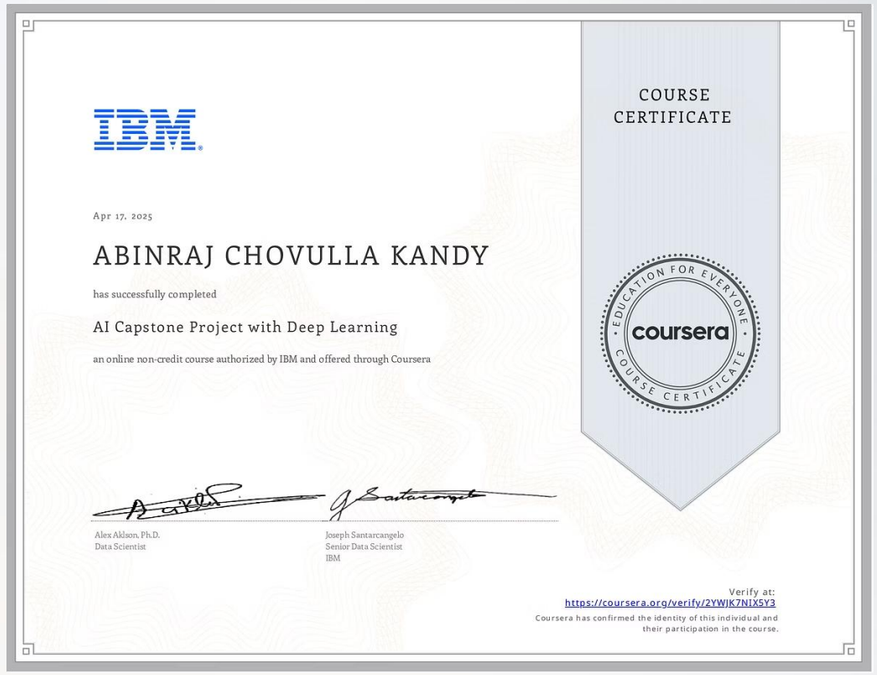
Transfer Learning

Use of ResNet and VGG models

Debugging Skills

Model tuning and optimization





Certificate Showcase

Official Recognition

IBM AI Engineering Professional Certificate

Project Completion

Successful deep learning capstone

Career Impact

Enhance AI and data science credibility

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Conclusion and Q&A

Practical Experience

Solidify AI skills through project work

Theory to Practice

Reinforce concepts with application

Open Floor

Engage with Q&A session

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